

TRANSPORT PHENOMENA II, CHE 312, SPRING 2025

FINAL EXAM

CLOSED BOOK, TIME: 120 MINUTES



The above schematic and the reference frame apply to both problems.

1. Liquid B is laminarly flowing between two flat walls placed a distance $2B$ apart, with the velocity profile

$$v_z = v_{z,max} \left[1 - \left(\frac{x}{B} \right)^2 \right] \quad v_z (z)$$

The solid walls are made of a species A that is slightly soluble in the liquid B . Species A diffuses into the liquid according to Fick's law for binary diffusion with a constant diffusivity D_{AB} , where it undergoes a chemical reaction according to a first order rate law $R_A = -k_1''' c_A$.

Answer the questions in terms of the following parameters (all assumed constant):

$v_{z,max}$: maximum velocity of the fluid in z -direction

B : half the distance between the two parallel walls

D_{AB} : Fick's law diffusivity of A in B

k_1''' : rate constant in units of inverse time.

- 1.1. By setting up a shell mole balance, derive a partial differential equation for the concentration of species A as a function of x and z . Assume the concentration of A is very small $x_A \ll 1$, and neglect the diffusive transport of A in z direction.

$$\int_V dV \frac{\partial}{\partial t} c_A = - \int_S dS \langle N_A | n \rangle + \int_V dV R_A$$

- 1.2. Sufficiently far downstream, the concentration profile c_A is independent of z . Find this asymptotic concentration profile as a function of x , assuming the concentration of A at both walls is c_{A0} .

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2. Very thick slurries and pastes sometimes move in a channel almost as a solid plug with a uniform velocity profile. Consider a fluid between two flat walls placed a distance $2B$ apart, with a uniform velocity profile $v_z = v_0$. For $z > 0$, heat is added to the fluid with a constant flux q_0 at both walls. The liquid enters the heated region of the slit at $z = 0$ with a uniform temperature T_0 .

Answer the questions in terms of the following parameters (all assumed constant):

v_0 : velocity of the fluid in z -direction

B : half the distance between the two parallel walls

k : thermal conductivity of the fluid

ρ : density of the fluid

c_p : isobaric heat capacity per unit mass of the fluid

q_0 : heat flux at the walls (assumed positive)

- 6/15 2.1. Write down a steady state macroscopic energy balance on the entire fluid between 0 and z to find the bulk average temperature as a function of z . Assume constant heat capacity and neglect the pressure dependence of enthalpy.

$$\int_V dV \frac{\partial}{\partial t} \rho \left(\frac{1}{2} v^2 + \phi + u \right) = - \int_S dS \rho \left(\frac{1}{2} v^2 + \phi + u \right) \langle v|n \rangle - \int_S dS \langle q|n \rangle - \int_S dS \langle v|\tau|n \rangle - \int_S dS p \langle v|n \rangle$$

$$T_b(z) \equiv \frac{\int_A dS v_z T}{\int_A dS v_z}$$

- 6/15 2.2. Write down a steady state shell energy balance to obtain a partial differential equation (PDE) for the temperature of the liquid as a function of x and z .

For large z , the temperature profile assumes the following asymptotic solution:

$$T(x, z) = T_b(z) + \psi(x)$$

- 9/15 2.3. Derive an ordinary differential equation (ODE) for $\psi(x)$ by replacing the asymptotic solution in the PDE derived in the previous part. Solve the ODE and evaluate the integration constants.

- 10/15 2.4. Using the asymptotic temperature profile derived above find the Nusselt number.

$$Nu = \frac{4hB}{k}$$

$$h[T(B, z) - T_b(z)] = q_0$$

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1.1

negligible

$$N_{Ax} = x_A (N_A + N_B) + J_{Ax}^* \approx J_{Ax}^* = -c D_{AB} \frac{dx_A}{dx}$$

Assuming c is constant: $J_{Ax}^* = -D_{AB} \frac{dC_A}{dx}$

negligible

$$N_{Az} = x_A (N_A + N_B) + J_{Az}^* \approx J_{Az}^* = -c D_{AB} \frac{dx_A}{dz}$$

Assuming c is constant: $J_{Az}^* = -D_{AB} \frac{dC_A}{dz}$

At S.S.

$$0 = [N_{Ax}]_x - [N_{Ax}]_{x+\Delta x} + [N_{Az}]_z - [N_{Az}]_{z+\Delta z} - k_1''' C_A$$

Taking $\lim_{\Delta x \Delta z \rightarrow 0}$:

$$0 = -\frac{\partial N_{Ax}}{\partial x} - \frac{\partial N_{Az}}{\partial z} - k_1''' C_A \quad \text{Neglecting } z \text{ direction} \Rightarrow 0 = -\frac{dN_{Ax}}{dx} - k_1''' C_A$$

1.2

Substituting in N_{Ax} expression

$$0 = D_{AB} \frac{d}{dx} \left(\frac{dC_A}{dx} \right) - k_1''' C_A$$

$$\Rightarrow \frac{d^2 C_A}{dx^2} = \frac{k_1''' C_A}{D_{AB}}, \quad \eta = \frac{k_1''' C_A}{D_{AB}}$$

$$\iint \frac{d^2 C_A}{dx^2} = \iint dx \quad \eta$$

$$C_A(x) = \frac{\eta x^2}{2} + C_1 x + C_2$$

B.C: ① $C_A(x = -B) = C_{A0}$

② $C_A(x = B) = C_{A0}$

Using B.C ②:

$$C_{A0} = \eta \frac{B^2}{2} + C_1 B + C_2$$

$$\Rightarrow C_{A0} = C_{A0} \Rightarrow 0 = 2C_1 B + 0$$

$$\Rightarrow C_1 = 0$$

$$C_{A0} = \eta \frac{B^2}{2} + C_2 \Rightarrow C_2 = C_{A0} - \eta \frac{B^2}{2}$$

$$\Rightarrow C_A(x) = \eta \frac{x^2}{2} + C_{A0} - \eta \frac{B^2}{2}$$

$$C_A(x) = C_{A0} + \frac{\eta}{2} (x^2 - B^2)$$

Using B.C ①:

$$C_{A0} = \eta \frac{B^2}{2} - C_1 B + C_2$$

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$$(2.1) \quad 0 = - \int_S dS \rho \left(\frac{1}{2} v^2 + \phi + u \right) \langle v | n \rangle - \int_S dS \langle q | n \rangle$$

$$(2.2) \quad 0 = - \int_S dS \langle q | n \rangle$$

$$0 = [q_x]_x - [q_x]_{x+\Delta x} + [q_z]_z - [q_z]_{z+\Delta z}$$

With $\lim_{\Delta x \Delta z \rightarrow 0}$:

$$0 = - \frac{\partial q_x}{\partial x} - \frac{\partial q_z}{\partial z} \quad \left| \quad q_x = -k \frac{\partial T}{\partial x}, \quad q_z = -k \frac{\partial T}{\partial z} \right.$$

$$\Rightarrow \boxed{0 = \frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right)}$$

Bulk temp in terms of z only:

$$\Rightarrow 0 = k \frac{d^2 T}{dz^2}$$

B.C: ① $T(z=0) = T_0$

② $\frac{dT}{dz} = q_0$

$$\int \left(\frac{d^2 T}{dz^2} \right) dz = \int 0 dz \quad 0$$

$$T(z) = C_1 z + C_2$$

Apply B.C ①:

$$T_0 = C_2$$

Apply B.C ②:

$$q_0 = C_1$$

$$\Rightarrow T(z) = q_0 z + T_0$$

$$T_b(z) = \frac{\int_A dS \, v_z T}{\int_A dS \, v_z}$$

$$= \frac{\int_{-B}^B dz \, (v_0) (q_0 z + T_0)}{\int_{-B}^B dz \, v_0}$$

$$= \frac{\int_{-B}^B dz \, q_0 v_0 z + T_0 v_0}{\int_{-B}^B dz \, v_0}$$

$$= \frac{\left[\frac{q_0 v_0 z^2}{2} + T_0 v_0 z \right]_{-B}^B}{\left[v_0 z \right]_{-B}^B}$$

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$$T_b(z) = \frac{\left(\cancel{q_0 \cancel{\nu_0} B^2} + T_0 \cancel{\nu_0} B \right) - \left(\cancel{\frac{q_0 \cancel{\nu_0} B^2}{2}} - T_0 \cancel{\nu_0} B \right)}{\cancel{\nu_0} B - (-\cancel{\nu_0} B)}$$

$$T_b(z) = \frac{2 T_0 \cancel{\nu_0} B}{\cancel{2 \nu_0} B}$$

$$T_b(z) = T_0$$

2.3 $T(x, z) = T_b(z) + \psi(x)$

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial z^2} = T_0 + \psi(x)$$

$$\rightarrow \psi(x) = \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial z^2} - T_0$$

2.4 $h = \frac{q_0}{T(B, z) - T_b(z)} = \frac{q_0}{T(B, z) - T_0}$

$$\rightarrow Nu = \frac{4 q_0 B}{k [T(B, z) - T_0]}$$

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